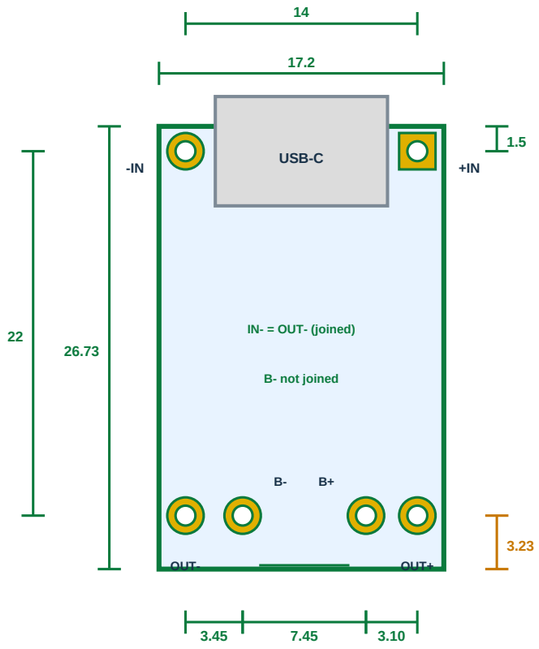
Solar Lights Lite Rev B: module geometry (board regenerated with these values)

Compare each drawing with the real module. GREEN = given by you / vendor drawing. AMBER = assumed or read off a drawing. RED = I need your answer.

1 TP4056 charger module

Vendor orientation: USB-C up, +IN / OUT+ on the right



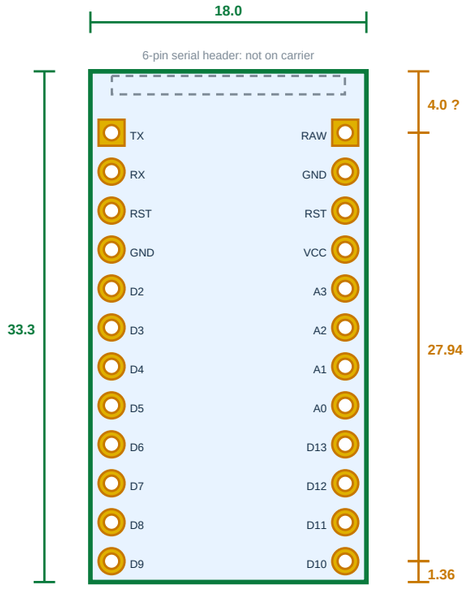
Your markup: length 26.73 (not 28), rows 22 apart (not 23), IN row 1.5, bottom 3.0.

7.7 mm = gap between the B- and B+ hole edges -> hole about 1.2 mm?

Copper pad size not given: 2.2 mm assumed (carrier side is unaffected).

2 Pro Mini (BTE13-010A clone)

Serial end up, as placed on the carrier. Only the two 12-pin rows are used



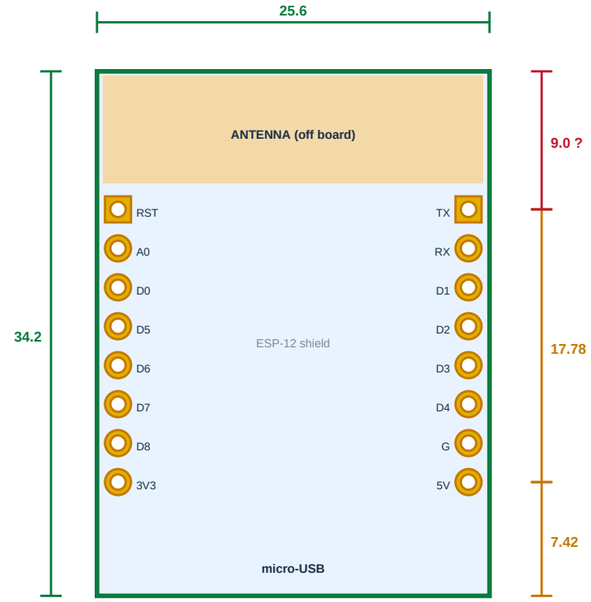
33.3 x 18.0, 27.94 span (2.54 pitch): ticked. Pin 1 at 4.0 from the serial end.

Photo of the real board gives 4.0 / 1.36 (sums to 33.3); 3.4 / 2.5 did not. Please re-check.

Row spacing 15.24 ticked. Template must be printed at exactly 100%.

3 Wemos D1 Mini

Antenna end up, as placed on the carrier

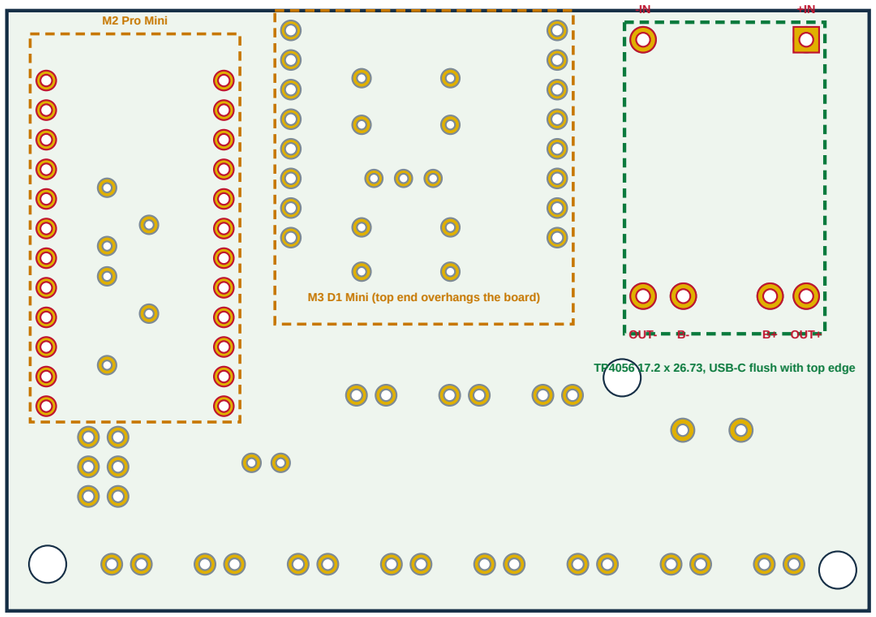


34.2 x 25.6: given. Row spacing 22.86 (0.9 in) is standard: assumed.

Pin 1 (RST, square) centre to antenna-end edge: please measure.

4 Board 74 x 51.5 mm with the new module sizes

Dashed = module bodies. Red = new TP4056 and Pro Mini holes. Grey = other holes on the current board.



What I will build from (after you confirm)

TP4056 hole centres

pad	u(from USB end)	v(from +side edge)	carrier x,y
+IN	1.5	1.60	68.60, 2.50
-IN	1.5	15.60	54.60, 2.50
OUT+	23.5	1.60	68.60, 24.50
B+	23.5	4.70	65.50, 24.50
B-	23.5	12.15	58.05, 24.50
OUT-	23.5	15.60	54.60, 24.50

IN- = OUT- on the module: no extra link on the carrier, JPV1 pad 2 is ground through the module.

B- not joined to OUT-: B- stays its own net on the carrier.

TP4056 17.2 x 26.73; Pro Mini 33.3 x 18.0 (pin 1 at 4.0 mm); D1 Mini 34.2 x 25.6; side sockets only.

Assumed (standard): Pro Mini rows 15.24; D1 Mini rows 22.86; pin pitch 2.54; TP4056 pad 2.2.

CHECK A Is your 7.7 mm the gap between the B- and B+ hole edges (so hole = 1.2 mm)?

CHECK B Pro Mini pin 1 at 4.0 / far end 1.36 (from your photo, +/-0.3): please confirm with calipers.

OPEN 3 D1 Mini: pin 1 (RST) centre to antenna-end edge (9.0 assumed). Not on your sheet.

OPEN 4 Socket height (8.5 mm assumed); parts under modules need about 6 mm clear.

On regenerate: Pro Mini pins move down 2.0 mm (pin 1 y 6.0); TP4056 body ends at y 27.7; silk shifts.